

PRASHAST VATS

Bangalore, India | +91 63631 40535 | prashastvats@gmail.com

[GitHub](#) | [Portfolio](#) | [Remora \(live\)](#)

Software engineer with 3+ years shipping end-to-end products, from enterprise .NET and React work for HP to independently built and deployed services running in production. Ships the whole slice: schema and API through to the interface, plus the deployment. Python and TypeScript first, with a habit of building the measurement layer alongside whatever I ship.

PROJECTS & INDEPENDENT WORK

Remora, Prompt-Injection Detection for AI Agents [remora.watch](#) · [Chrome Web Store](#) · [Repo](#) | 2026

- Published Chrome extension detecting hidden prompt-injection payloads aimed at AI agents. Eight structural detectors cover CSS-concealed text, off-screen positioning, computed color-match, aria-hidden nodes, comment nodes, and `::before/::after` injection.
- Dual-signal design to control false positives:** a structural hit alone never fires; extracted text must also match one of 25 injection patterns across 5 categories. Tuned to **zero false positives** across seven major production sites, with a 49-case suite carrying explicit false-positive fixtures.
- End to end solo in **3 weekends**: TypeScript detection engine, MV3 content script, service worker, Shadow DOM popup UI, and build pipeline. Sub-100ms synchronous detection path. **49 installs at ~90% retention**, zero promotion; submitted to Y Combinator (F2026).

AI Agent Auth Gateway [Live Demo](#) · [Backend](#) · [Frontend](#) | 2026

- Open-source delegated-authority system where an agent acts on behalf of a user while retaining its own identity. Implements **RFC 8693 token exchange**, minting short-lived delegation JWTs that carry both identities.
- Policy middleware** enforcing per-agent endpoint, method, and time-window rules with a full **audit trail**, plus a human-in-the-loop consent gate on destructive operations. React + Tailwind dashboard over agents, policies, consent approvals, and logs, with OIDC Authorization Code + PKCE login.
- Stack:** Python · FastAPI · PostgreSQL (Neon) · SQLAlchemy · React · Vite · Tailwind · Render/Vercel.

Priya, Autonomous Outbound Voice Agent [Live Demo](#) | Winner, Voice Agent Jam 3.0 · May 2026

- Voice agent placing **real outbound calls over SIP/PSTN** that handles pricing, competitor, and interest objections in code-switched Hindi/English. Built solo and won against team entries.
- FastAPI backend orchestrating Retell AI and Vobiz telephony, with a React dashboard scoring enrollment rate, objection breakdown, and per-call sentiment. **Stack:** Python · FastAPI · Retell AI · SIP/PSTN · React · Render.

Event Intelligence Dashboard [Live Demo](#) | 2026

- LLM-backed service aggregating multi-source news into running summaries for user-defined events. **Stack:** FastAPI · APScheduler · Gemini · Appwrite · React/Vite · Vercel.

PROFESSIONAL EXPERIENCE

Software Engineer · Mphasis Ltd. (HP Inc. engagement) | May 2023 to Present

Full-stack delivery, test automation, and AI tooling for HP's global printing division.

- Built a **ResNet + FAISS computer-vision service** identifying printer UI elements from screenshots, replacing a paid Azure AI Vision dependency and cutting recurring cost. Containerised with Docker and Docker Compose for Windows and Linux.
- Built a **.NET MVC web application** end to end, frontend and backend, that scans networked printers over SNMP OIDs, visualises fleet state, and automates firmware upgrades with Selenium.
- Developed an **AI onboarding assistant** on Copilot for HP's automation framework, covering script explanation, syntax guidance, and code generation. Placed 2nd of 13 developers in the internal hackathon.
- Embedded with **HP's Spain engineering team**, building automation and MariaDB-backed analytics. Contributed to Mphasis' first European engagement, which converted into a continued partnership.
- Resolved **15+ tickets monthly at ~80% first-contact resolution**, supporting **30+ QA engineers across 3 time zones**.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL

Frontend: React, React Native, Vite, Tailwind CSS, Angular, HTML5/CSS3, Chrome Extensions (Manifest V3)

Backend: FastAPI, .NET MVC, ADO.NET, REST API design, middleware, background jobs, webhook handling

Data: PostgreSQL, Microsoft SQL Server, MariaDB, MongoDB, SQLAlchemy, Appwrite

AI/LLM: OpenAI/Claude/Gemini APIs, prompt engineering, LLM output evaluation, RAG, FAISS, computer vision (ResNet), voice agents (Retell AI, SIP/PSTN)

Auth & Infra: OAuth 2.0 / OIDC, JWT, PKCE, Docker & Docker Compose, Jenkins CI/CD, Linux, Render, Vercel, Git/GitHub

EDUCATION & RECOGNITION

B.Tech, Electronics & Instrumentation Engineering · Manipal Institute of Technology | 2018 to 2022

- **Winner, Voice Agent Jam 3.0** (Bangalore, 2026), competing and shipping solo against team entries.
- **Project Shipyard S2**, an AI Learn Circle cohort supported by Venture Catalysts, Antler, Bhive, and Paytm.
- **Certifications:** Mphasis TalentNext (React.js, Angular, TypeScript, SQL, Git, GitHub Copilot, Agile) · Interactive Python, UX Design (Coursera).